Listing 4.33 Indexing a consultation document with doctor notes

```
PUT consultations/_doc/1
{
  "patient_name":"John Doe",
  "doctor_notes":{
    "temperature":103,
    "symptoms":["chills","fever","headache"],
    "history":"none",
    "medication":["Antibiotics","Paracetamol"]
  }
}
```

The flattened field can hold any number of subfields.

All of these fields are indexed as keywords.

As you can see, `doctor_notes` holds a lot of information—but remember, we did not create these inner fields in our mapping definition. Because `doctor_notes` is a `flattened` type, all the values are indexed as is, as keywords.

Finally, we can search the index using any keywords from the doctor notes.

Listing 4.34 Searching the `flattened` data type field

```
GET consultations/_search
{
  "query": {
    "match": {
      "doctor_notes": "Paracetamol"
    }
  }
}
```

Searching for a patient's medication

Searching for `"Paracetamol"` returns John Doe's consultation document. You can experiment by changing the `match` query to any of the fields—for example, `doctor_notes:chills`—or write a complex query like the one shown next.

Listing 4.35 Advanced query on a `flattened` data type

```
GET consultations/_search
{
  "query": {
    "bool": {
      "must": [
         {"match": {"doctor_notes": "headache"}},
         {"match": {"doctor_notes": "Antibiotics"}}
       ],
       "must_not": [
         {"term": {"doctor_notes": {"value": "diabetes"}}}
       ]
    }
  }
}
```